

Software Requirements Specification

for

Startup Incubator Management System

Version 1.0

Prepared by
Group No:08

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Chapter 1

Introduction

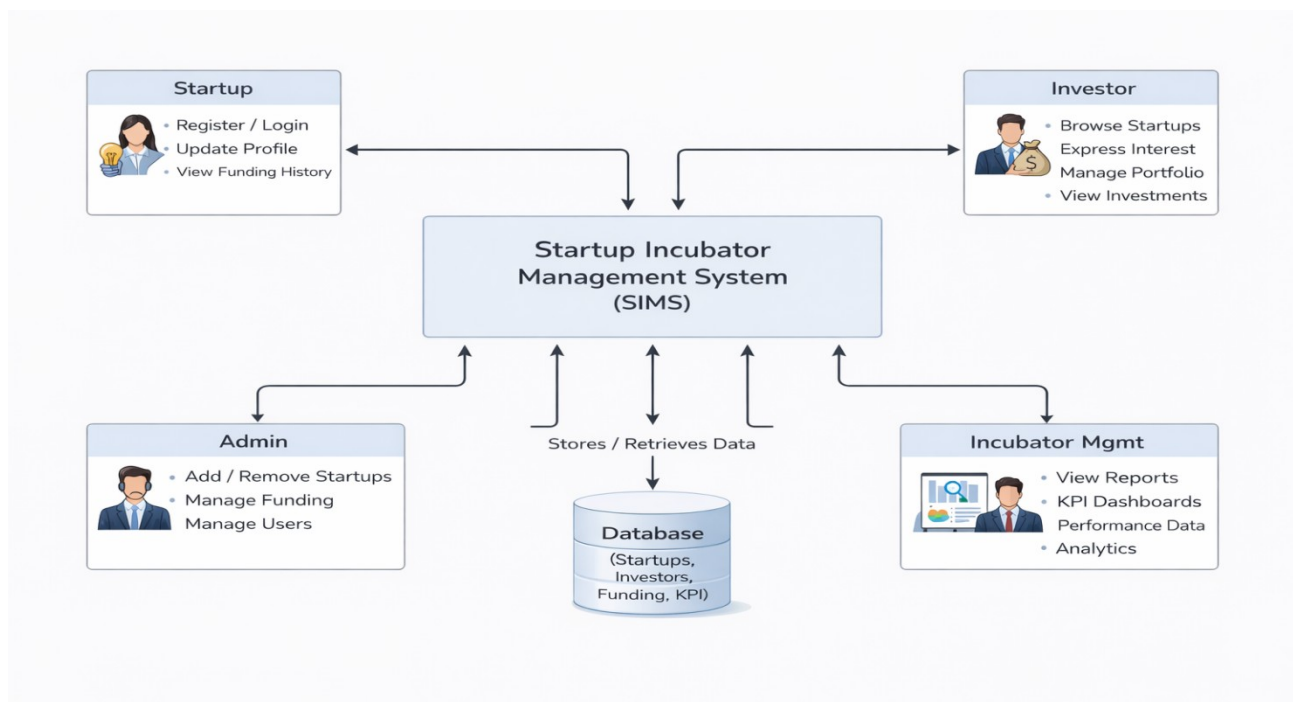
This project is a prototype for a Startup Incubator Management System (SIMS), a full-fledged database application designed to manage and streamline all operations within a startup incubator ecosystem. The system is intended to handle interactions between the incubator management, investors, startups, and other stakeholders involved in the pipeline. This section provides an overview of the purpose, scope, intended audience, and conventions used throughout this document.

1.1 Document Purpose

The main purpose of this document is to provide a detailed and comprehensive analysis of the Startup Incubator Management System (SIMS). This Software Requirements Specification (SRS) reflects both the functional and non-functional requirements needed for achieving the objectives of the product. The document also provides clarity on the external interface requirements, use case models, and system constraints. It serves as the baseline for system design, development, testing, and maintenance.

1.2 Product Scope

The Startup Incubator Management System provides a centralized digital platform for the incubator to manage startups, investors, funding rounds, and associated resources. The system will replace manual or fragmented data management practices, enabling efficient coordination among all stakeholders. Key features include startup registration and tracking, investor portfolio management, funding pipeline oversight, and report generation. The system is designed to be scalable and adaptable to various incubator environments.



1.3 Intended Audience and Document Overview

The application is intended to be used by system administrators working in the startup incubator, investors, startups, and incubator management. The types of users of the database are categorized as:

- Casual User → Incubator Management (views dashboards and reports)
- Naive User → System Administrators (manages data entry and updates)
- Standalone Users → Investors, Directors (accesses specific modules relevant to their role)

1.4 Definitions, Terms and Abbreviations

S.No.	Term	Definition
1	SIMS	Startup Incubator Management System
2	SRS	Software Requirements Specification
3	DBMS	Database Management System
4	Admin	System Administrator with full database access
5	Startup	An early-stage company enrolled in the incubator program
6	Investor	Individual or entity providing funding to startups
7	KPI	Key Performance Indicator
8	MOU	Memorandum of Understanding
9	API	Application Programming Interface

1.5 Document Conventions

This document follows the IEEE Std 830-1998 formatting requirements for Software Requirements Specifications. Functional requirements are labeled F1, F2, ..., Fn. Use cases are labeled U1, U2, ..., Un. All requirements use "shall" to indicate mandatory behavior.

1.6 References and Acknowledgments

- Fundamentals of Database Systems by Ramez Elmasri and Shamkant Navathe
- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications
- IEEE SRS Template – <http://www.cse.msu.edu/~cse870/IEEEExplore-SRS-template.pdf>

Chapter 2

Overall Description

2.1 Product Overview

The chosen mini world for this project is that of a startup incubator. A business incubator is a company that helps new startup companies to develop by providing services. The incubator deals at both ends — with the investors as well as the startups. The SIMS will serve as the central data management platform for this incubator. It will store and manage data relating to startups, investors, funding rounds, and milestones. The system will maintain a comprehensive record of all activities and transactions occurring within the incubator.

The system administrator will act as the super admin, having direct access to the database to add, delete, and update records.

2.2 Product Functionality

The SIMS will support the following core functionalities:

The system will allow:

- Any startup or investor to register into the system with their relevant details.
- Registered users to log in with valid credentials and access role-specific dashboards.
- Startups to update their profiles, milestones, and funding requirements.
- Investors to browse startup profiles, express interest, and manage their portfolios.
- Automated tracking of funding rounds, MOU signing, and equity details.
- Generation of progress reports and KPI dashboards for management.
- Notification alerts for upcoming deadlines, funding milestones, and review meetings.

2.3 Design and Implementation Constraints

All changes to startup profiles and investment deals must be reflected in real time in the database. Financial data such as investment amounts must be stored with high precision. The system will be accessible via any standard web browser with an internet connection. Users must authenticate with valid credentials before accessing any module. Data stored must comply with applicable data protection and privacy regulations.

2.4 Assumptions and Dependencies

The following assumptions are made:

- All startup, investor data is stored in a relational database.
- The system will have sufficient storage and provide fast access to data.
- The system will be available 24/7 with minimal downtime.
- Users are assumed to have basic digital literacy, the interface will be in English.

The following dependencies are noted:

- The hardware and software described in Section 3.1.2 and 3.1.3.
- The system administrator must have a working knowledge of incubator operations.
- The database must be updated upon every transaction (registration, funding, milestone update, etc.).

Chapter 3

Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

- The application opens to a login/registration page upon launch.
- Role-based dashboards are shown post-login (Admin, Investor, Startup).
- The interface uses a clean, minimal layout with intuitive navigation menus.
- Colour schemes are chosen to be accessible and readable.
- All forms include validation with descriptive error messages.
- Search and filter functionalities are available across all major data views.

3.1.2 Hardware Interfaces

- Supported Devices: The system will be accessible through web browsers, making it compatible with desktop ,laptops, tablets, and mobile devices.
- Input Devices: Users will interact with the system using keyboards, mice, or touchscreens.
- Servers: The backend will run on a cloud or a local server, processing requests and handling database operations.
- Network Interfaces: The system will communicate over the internet via HTTPS.

3.1.3 Software Interfaces

Software Used	Description
Operating System	Linux/Windows – chosen for broad deployment support and compatibility.
Database	MySQL – an open-source relational DBMS used to manage and organize the database of the SIMS
Backend Framework	Flask (Python) – a lightweight web framework for building RESTful APIs.
Frontend	React.js with HTML & CSS – for building interactive, component-driven user interfaces.
Version Control	Git / GitHub – for collaborative development and code version management.

3.2 Functional Requirements

F1: The system shall allow a startup representative or investor to register into the system by providing their details. Registration data shall be stored in the database.

F2: The system shall allow registered users to log in using valid credentials. Role-based access shall be enforced upon successful login.

F3: The system shall deny access and display an error message 'Invalid Credentials' if incorrect login details are provided.

F4: The system shall allow administrators to add new startups to the incubator program with all relevant details (name, domain, founding date etc.).

F5: The system shall allow administrators to update startup profiles, including milestone completions, funding status, and team changes.

F6: The system shall allow administrators to remove a startup from the system when it graduates or exits the program.

F7: The system shall allow investors to view all registered startups and filter them by domain, funding stage, and performance metrics.

F8: The system shall allow investors to express interest in a startup and initiate the investment discussion pipeline.

F9: The system shall allow administrators to record and manage funding rounds including amount, equity percentage, investor details, and MOU status.

F10: The system shall allow startups to view their funding history, and milestone tracker.

F11: The system shall allow administrators to generate progress reports for individual startups.

F12: The system shall send automated notifications to relevant users regarding upcoming review meetings, funding deadlines, and milestone due dates.

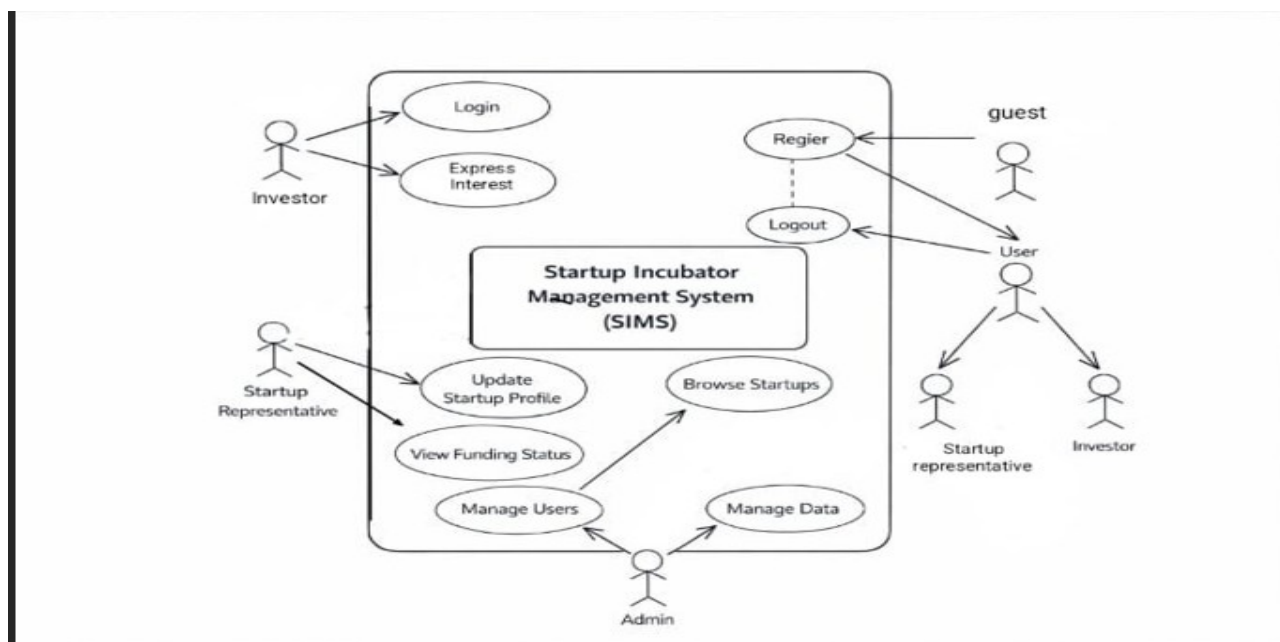
F13: The system shall allow administrators to view the complete activity log and audit trail for all data changes.

F14: The system shall allow investors to view their investment portfolio, including returns data and equity held across startups.

F15: The system shall provide a search functionality for all entities (startups, investors) across the system.

F16: The system shall enforce role-based access control so that users can only perform actions permitted for their role.

3.3 Use Case Model



3.3.1 Login – U1

Author	Rasvitha
Purpose	To log in to the system with valid credentials.
Requirements Traceability	F2, F3
Priority	High
Preconditions	The user must have a registered account with valid credentials.
Post Conditions	The user is successfully logged into their role-specific dashboard.
Actors	Admin, Investor, Startup Representative
Basic Flow	1. User enters credentials. 2. System verifies. 3. User is redirected to dashboard.
Alternative Flow	User enters incorrect credentials. System shows 'Invalid Credentials' error. User is prompted to retry or register.
Exceptions	Account locked after repeated failed attempts.

3.3.2 Register – U2

Author	Varshita
Purpose	To register a new user (startup or investor) into the system.
Requirements Traceability	F1
Priority	High
Preconditions	The user must not already be registered in the system.
Post Conditions	User account is created and credentials are stored in the database.
Actors	Startup Representative, Investor
Basic Flow	1. User fills registration form. 2. System validates inputs. 3. Account is created and stored.
Alternative Flow	If the user already exists, the system shows a message and redirects to the login page.
Exceptions	Submission of incomplete or invalid registration details.

3.3.3 Add Startup – U3

Author	Richa
Purpose	To add a new startup to the incubator program.
Requirements Traceability	F4
Priority	High
Preconditions	Admin must be logged in. Startup details must be complete and verified.
Post Conditions	Startup is added to the database and appears in the system.
Actors	Startup Representative
Basic Flow	1. Startup Representative fills in startup details. 2. System validates. 3. Startup record is created in the database.
Alternative Flow	If a startup with the same name already exists, the system alerts the admin.
Exceptions	Submission of incomplete mandatory fields.

3.3.4 Update Startup Profile – U4

Author	Sai Roshini
Purpose	To update the profile, milestones, or funding status of an existing startup.
Requirements Traceability	F5
Priority	High
Preconditions	Admin or Startup Representative must be logged in. Startup must exist in the system.
Post Conditions	Updated information is reflected in the database.
Actors	Admin, Startup Representative
Basic Flow	1. User selects startup profile. 2. Edits relevant fields. 3. System saves changes to the database.
Alternative Flow	If the user edits a field with invalid data, an inline error message is shown.
Exceptions	Concurrent modification conflicts if two users edit the same record.

3.3.5 Browse Startups (Investor) – U5

Author	Rasvitha
Purpose	To allow investors to browse and filter startup profiles.
Requirements Traceability	F7
Priority	High
Preconditions	Investor must be logged in.
Post Conditions	A filtered list of startups matching the investor's criteria is displayed.
Actors	Investor, Startup Representative
Basic Flow	1. They open startup directory. 2. Applies filters (domain, stage, etc.). 3. Filtered results are displayed.
Alternative Flow	If no startups match the filter, a 'No results found' message is shown.
Exceptions	Database connection failure.

3.3.6 Express Interest – U6

Author	Varshita
Purpose	To allow an investor to express interest in a startup and initiate the investment pipeline.
Requirements Traceability	F8
Priority	Medium
Preconditions	Investor must be logged in. Target startup must exist in the system.
Post Conditions	Interest is logged and admin is notified for follow-up.
Actors	Investor
Basic Flow	1. Investor views startup profile. 2. Clicks 'Express Interest'. 3. System logs the action and notifies admin.
Alternative Flow	If the investor has already expressed interest, the system shows a notification that interest is already recorded.
Exceptions	Investor expresses interest in a startup that has already closed its funding round.

3.3.7 Record Funding Round – U7

Author	Richa
Purpose	To record the details of a new funding round for a startup.
Requirements Traceability	F9
Priority	High
Preconditions	Admin must be logged in. Startup and investor must both be registered in the system.
Post Conditions	Funding round is recorded in the database and startup funding status is updated.
Actors	Admin
Basic Flow	1. Admin selects startup. 2. Enters funding round details (amount, equity, investor, MOU status). 3. System saves the record.
Alternative Flow	If the equity percentage makes the total exceed 100%, the system shows a validation warning.
Exceptions	Admin records duplicate funding rounds for the same startup and investor.

3.3.8 Generate Report – U10

Author	Sai Roshini
Purpose	To generate progress reports for startups.
Requirements Traceability	F11
Priority	Medium
Preconditions	Admin must be logged in.
Post Conditions	A report is generated and made available for download.
Actors	Admin
Basic Flow	1. Admin selects report type (individual startup). 2. System compiles relevant data. 3. Report is displayed and offered for download (PDF/CSV).
Alternative Flow	If no data is available for the selected period, a 'No data found' message is shown.
Exceptions	Admin requests a report for a startup that no longer exists in the system.

3.3.9 View Portfolio (Investor) – U11

Author	Rasvitha
Purpose	To allow an investor to view their investment portfolio.
Requirements Traceability	F14
Priority	Medium
Preconditions	Investor must be logged in.
Post Conditions	Portfolio details including equity percentages and performance metrics are displayed.
Actors	Investor
Basic Flow	1. Investor navigates to Portfolio section. 2. System fetches all investment records. 3. Portfolio is displayed with aggregated statistics.
Alternative Flow	If no investments exist, a message prompts the investor to explore startups.
Exceptions	Portfolio contains data for a startup that has been archived.

3.3.10 Notifications – U12

Author	Varshita
Purpose	To send automated notifications to users for upcoming events and deadlines.
Requirements Traceability	F12
Priority	Medium
Preconditions	The triggering event (meeting, deadline, milestone) must be scheduled in the system.
Post Conditions	Notification is delivered to relevant users.
Actors	System (automated), Admin, Startup Representative, Investor
Basic Flow	1. System detects an upcoming event. 2. Generates notification. 3. Notification is sent to the user's dashboard.
Alternative Flow	If user email is not verified, notification is sent only to the in-app dashboard.
Exceptions	System fails to send notification due to connectivity issues.

Chapter 4

Other Non-Functional Requirements

4.1 Performance Requirements

The system is designed to handle lightweight queries, ensuring quick data retrieval and minimal server load. UI transitions and screen changes will be smooth, requiring minimal computation. Server updates should complete within a few seconds, provided a stable internet connection is maintained.

4.2 Safety and Security Requirements

All sensitive financial and personal data shall be encrypted both at rest and in transit using industry-standard protocols. Role-based access control shall be strictly enforced so that users can only access data relevant to their role. Regular automated database backups shall be performed.

4.3 Software Quality Attributes

4.3.1 Reliability

The system shall maintain 99% uptime. Failure recovery mechanisms shall ensure no data loss in case of unexpected shutdowns.

4.3.2 Usability

The interface shall be intuitive and usable without formal training for all user roles. A help/documentation section shall be accessible from every page.

4.3.3 Maintainability

The codebase shall follow standard software engineering practices (modularization, documentation) to allow easy maintenance and future enhancements.

4.3.4 Portability

The system shall be deployable on any standard server infrastructure. The frontend shall be compatible with modern browsers (Chrome, Firefox, Safari, Edge).

4.3.5 Scalability

The system architecture shall support horizontal scaling to accommodate future growth in the number of startups, investors, and transactions.

Appendix A – Data Dictionary

Item Name	Possible Values / States	Description	Related Operations / Requirements
User_ID	Unique Integer	Unique identifier for every user in the system	Used in login, authentication, and user management
User_Role	Admin, Investor, Startup Representative	Defines the access privileges of the user	Enforced by Role-Based Access Control (F16)
Startup_ID	Unique Integer	Unique identifier assigned to each startup	Used for tracking startups and linking funding records
Investor_ID	Unique Integer	Unique identifier for investors	Used to manage portfolios and investments
Funding_Round_ID	Unique Integer	Identifier for each funding round event	Used in funding round tracking and reporting
Startup_Status	Active, Graduated, Exited	Represents the current status of a startup in the incubator	Updated by Admin during life cycle events (F4–F6)
Funding_Status	Not Funded, Seed, Series A, Series B	Represents the stage of funding for a startup	Updated during funding round recording (F9)
MOU_Status	Pending, Signed, Cancelled	Indicates whether a Memorandum of Understanding is finalized	Managed during investment processing
Registration_Form	Name, Email, Password, Role, Startup/Investor Details	Data entered by users when registering for the system	Used in user registration (F1)
Login_Credentials	Username, Password	Used to authenticate users	Used in login verification (F2, F3)
Startup_Profile_Data	Startup Name, Domain, Founding Date, Team Details	Information used to create or update startup profiles	Managed by Admin or Startup Representative (F4, F5)
Funding_Details	Investment Amount, Equity %, Investor ID, MOU Status	Details recorded during funding rounds	Used in funding management (F9)
Search_Query	Keyword, Domain Filter, Funding Stage Filter	Allows investors/admin to search startups or investors	Used in system-wide search functionality (F15)
Dashboard_View	Role-based data display	Displays relevant data depending on user role	Generated after login
Startup_List	List of startups with filters applied	Displays startups to investors/admin	Used in browsing startups (F7)
Portfolio_View	Investments, equity percentage, returns	Displays investor's portfolio	Used in portfolio management (F14)
Notification_Message	Deadline alerts, milestone reminders	Automated notifications sent to users	Used in notification system (F12)

Term	Description
Admin / System Administrator	Super user with full CRUD access to all entities in the database. Manages all system operations.
Startup	An early-stage company enrolled in the incubator program. Has a profile with team details, domain, milestones, and funding history.
Investor	An individual or entity that provides financial capital to startups. Has a portfolio view and can interact with the startup pipeline.
Funding Round	A structured investment event wherein an investor provides capital to a startup in exchange for equity or convertible notes.
Milestone	A defined goal or achievement target set for a startup to track progress within the incubator program.
MOU	Memorandum of Understanding – a formal agreement between the incubator/investor and the startup outlining terms of engagement.
KPI	Key Performance Indicator – measurable values used to evaluate the progress and success of a startup.
Portfolio	The collection of startups in which an investor has made investments, viewable from the investor dashboard.

Appendix B - Group Log

Meeting Day	TimeLine
12/03/2026	6:00-8:00PM
13/03/2026	8:00-9:30PM
14/03/2026	3:00-5:00PM

12 March 2026

This was the first meeting of our project group where we discussed the problem domain and finalized the topic Startup Incubator Management System (SIMS). We discussed how incubators manage startups and investors and identified the need for a centralized system to track startups, investments, funding rounds, and milestones. During this meeting, we also started outlining the structure of our Software Requirements Specification (SRS) document and divided the sections among team members.

13 March 2026

In this meeting, we focused on defining the system requirements and use cases. We discussed the major entities required in the system such as Users, Startups, Investors, Funding Rounds, Milestones, and Reports. We drafted the functional requirements and began designing the Use Case Model. We also finalized the technology stack, deciding to use MySQL as the database, Flask (Python) for the backend, and React.js with HTML/CSS for the frontend.

14 March 2026

In the final meeting, we reviewed and refined the SRS document. Each member proofread the sections they were responsible for and suggested improvements where necessary. We finalized the functional requirements, use case descriptions, and non-functional requirements. The document was formatted according to the IEEE SRS standard, and the final version was prepared for submission.